

Technical Summary

Customer Support Q&A; API | Node.js implementation notes

Architecture

- Express provides the REST API and JSON request parsing.
- The /support/ask endpoint validates the incoming customer question.
- Prompt helpers combine a system prompt with a structured support context.
- OpenAI Chat Completions returns the customer-facing answer.
- Errors are mapped into clear API responses: 400, 429, 502, and 504.
- Core validation and prompt logic can be tested without an API key.

Files Included

File	Purpose
src/server.js	Express API routes and response handling
src/openaiClient.js	OpenAI client and answer generation
src/prompt.js	System and user prompt builders
src/validation.js	Request validation and error mapping
src/supportContext.js	Sample business support context
test/run-tests.js	Small tests for core logic

Adaptation Options

- Replace sample support context with a client FAQ or help center content.
- Connect the endpoint to a website chat widget or contact form.
- Return structured JSON with escalation flags, categories, or suggested next actions.
- Add logging to Google Sheets, Airtable, or a CRM.
- Deploy to Render, Railway, Fly.io, Vercel serverless functions, or a client server.